

Problem 3

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Problem 1 The limit of Riemann sums for a function f on the interval $[1, 5]$ is given by

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left(2x_k^* + \frac{1}{x_k^*} \right) \Delta x \text{ on } [1, 5].$$

(a) Identify f and express the limit as a definite integral.

Explanation.

$$f(x) = 2x + \frac{1}{x}$$

$$\int_1^5 \left(2x + \frac{1}{x} \right) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left(2x_k^* + \frac{1}{x_k^*} \right) \Delta x$$

(b) Evaluate the limit of Riemann sums.

Explanation.

$$\int_1^5 \left(2x + \frac{1}{x} \right) dx = \left(x^2 + \ln |x| \right) \Big|_1^5 = (25 + \ln(5)) - (1 + \ln(1)) = 24 + \ln(5)$$

[4]